

Varun Srivastava

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EDUCATION

University of California, Berkeley

B.A. Computer Science, Mathematics

Graduated Dec 2023

GPA: 3.9/4

EXPERIENCE

Machine Learning Engineer

AdMarketplace

August 2024 – Present

New York, NY

- Playing a key role in implementing vector search for our full database of products (> 100 million titles) by overseeing embedding fine-tuning and benchmarking, improving relevance.
- Designing and training ML time-series models for the firm's algorithmic ad bidding solution, optimizing millions of dollars in daily ad impressions.
- Built a full-stack prototype for an AI shopping assistant; implemented an end-to-end RAG pipeline with embedding-based retrieval and LLM generation.

Founding Engineer

Stork Oracle Network

August 2022 – December 2022

San Francisco, CA

- Designed a high-speed and reliable WebSocket API for live crypto data. The pricefeed served several notable paying DeFi clients, such as Vertex and Apex.
- Scaled the full-stack Python infrastructure up to meet high uptime and low latency requirements for platforms trading millions of dollars daily.

Quantitative Analysis Intern

Dexterity Capital

May 2022 – August 2022

Seattle, WA

- Implemented CMA-ES strategy in forecasting models, improving hyperparameter convergence by > 100x.
- Analyzed signals in cryptocurrency markets that became incorporated in the company's proprietary high frequency trading (HFT) market making algorithm, which trades billions of dollars in volume a day.

Data Science Intern

First Republic Bank

June 2020 – August 2020

San Francisco, CA

- Optimized the internal data orchestration (CI/CD) pipeline by designing an Apache Airflow pipeline with scheduled tasks and dependencies, decreasing end-to-end execution time of tasks by 50%.

RESEARCH & PUBLICATIONS

Machine Learning & Vision Researcher | *Berkeley Visual Computing Lab*

December 2023 – May 2024

- Lee, J., **Srivastava, V.**, Jennings, N., Ng, R. "Theory of Human Tetrachromatic Color Experience." *ACM SIGGRAPH*, 2024. Honorable Mention for Technical Papers Awards, spotlighted at the annual conference.
- Researched the emergence of color vision in humans via neurological models, specifically in the case of esoteric conditions such as human tetrachromacy.

Vision & Neuroscience Researcher | *Na-Ji Biophysics Lab*

September 2019 – April 2020

- Implemented computer vision tools in Matlab for researchers to track the pupil size of rats in video data.
- Correlated terabytes of time-series data with pupil measurements to analyze how rodents react to shock stimulus.

PROJECTS

Modded NanoGPT | <https://github.com/KellerJordan/modded-nanogpt>

- Achieved multiple world records in a popular LLM training competition (> 3200 Github stars).

Spotify MCP | github.com/varunneal/spotify-mcp

- MCP Server integrating Spotify with LLMs (> 500 GitHub stars). Winner at the Anthropic MCP Hackathon.

Semantic Finder | do-me.github.io/SemanticFinder

- Developed a semantic search site powered by LLMs and a corresponding Chrome extension (> 300 GitHub stars).

TECHNICAL SKILLS

Languages: Python, Triton, C/C++, JavaScript, Go, Java, SQL, Matlab.

Frameworks & development: AWS, React, Jenkins, Docker, Bitbucket, Kubernetes Node, Django.

Data & ML: PyTorch, CUDA, Huggingface, AWS Sagemaker.